CMU-CS 462: Software Measurement and Analysis

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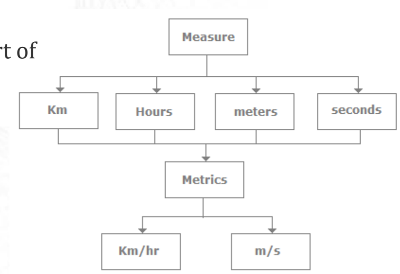
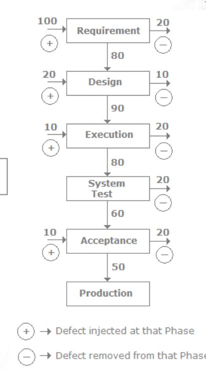
Review

Size and Complexity are a part of
A. Product Metrics
B. Process Metrics
C. Project Metrics
D. None of the above

Cost and Schedule are a part of
A. Product Metrics
B. Process Metrics
C. Project Metrics
D. None of the above





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Review

Which of the following does not affect the software quality and organizational performance?

- A Market
- B Product
- C Technology
- D People

Which of the following is not a direct measure of SE process?

- A Efficiency
- B Cost
- C Effort Applied
- D All of the mentioned

The intent of project metrics is:

- A minimization of development schedule
- B for strategic purposes
- C assessing project quality on ongoing basis
- D minimization of development schedule and assessing project quality on ongoing basis

Which of the following is an indirect measure of product?

- A Quality
- B Complexity
- C Reliability
- D All of the Mentioned

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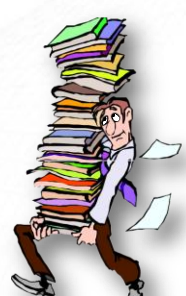
Contents

- Software measure classification
- Goal-based paradigms:
  - Goal-Question-Metrics (GQM)
  - Goal-Question-Indicator-Metrics (GQIM)
- Applications of GQM and GQIM



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Goal-Based Measurement (GBM)

- The primary question in goal-based measurement:
- "What do we want to know or learn?"
  instead of "What metrics should we use?"
- Because the answers depend on your goals, no fixed set of metrics is universally appropriate.
- Instead of attempting to develop general-purpose measures, one has to describe an adaptable process that users can use to identify and define measures that provide insights into their own development problem.

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GBM Process

- **Determining what to measure**
  - Identifying entities
  - Classifying entries to be examined
  - Determining relevant goals
- **Determining how to measure**
  - Inquire about metrics
  - Assign metrics

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Identifying Entities

- Product, process, resource [Fenton' 91].
- Artifacts, activities, agents [Armitage' 94].
- Both schemes are incomplete and haven't enough representational power.
- E.g., neither seems to deal with entities such as "defects".
  - Is a defect a product, a process, or a resource?
  - Is defect an artifact, activity, or agent?

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Process, Product, Resource

- **Process**
  - A collection of software related activities usually associated with some timescale.
  - Different SE processes: development, maintenance, testing, reuse, configuration and management process, etc.
- **Product**
  - Any artifacts, deliverables or documents that result from a process activity.
- **Resource**
  - Entities required by a process activity.

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Types of Attributes

- **Internal attributes**
  - Attributes that can be measured entirely in terms of the process, product or resource itself separate from its behaviour ← e.g., size
- **External attributes**
  - Attributes that can be measured with respect to how the process, product or resource relates to its environment through its behaviour
    ← e.g., quality

Sample Process Measures/1

Process Entities	Attributes	Possible Measures
development process	elapsed time	Calendar days, working days
	milestones	Calendar dates
	development effort	Staff-hours, days, or months
	phase containment	Percent of total defects found in phase where introduced
	process compliance	Percent of tasks complying with standard procedures or directives
	performance	Number of tests passed divided by number of tests executed
test process	volume	Number of tests scheduled
	progress	Number of tests executed Number of tests passed

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Sample Process Measures/2

Process Entities	Attributes	Possible Measures
detailed design	elapsed time	calendar days, working days
	design quality	defect density: number of design defects found in down-stream activities divided by a measure of product size, such as function points or physical source lines of code.
maintenance	cost	dollars per year staff-hours per change request
Change request backlog	size	number of change requests awaiting service estimated effort (staff-hours) for pending requests

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Sample Product Measures/1

Product Entities	Attributes	Possible Measures
system	**size**	**number of modules** **number of bubbles in a data-flow diagram number of function points** **number of physical source lines of code** **number of memory bytes or words required (or allocated)**
	defect density	**defects per KLOC** **defects per function point**
module	**length**	**physical source lines of code logical source statements**
	percent reused	**ratio of unchanged physical lines to total physical lines, comments and blanks excluded**

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Sample Product Measures /2

Product Entities	Attributes	Possible Measures
unit	**number of linearly independent flowpaths**	**McCabe's complexity**
document	**length**	**number of pages**
line of code	**statement type**	**type names**
	how produced	**name of production method**
	programming language	**language name**
defect	**type**	**type names**
	origin	**name of activity where introduced**
	severity	**an ordered set of severity classes**
	effort to fix	**staff-hours**

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Sample Resource Measures

Resource Entities	Attributes	Possible Measures
Assigned staff	team size	number of people assigned
	experience	years of domain experience years of programming experience
CASE tools	type	name of type
	Is used?	yes/no (a binary classification)
time	start date, due date	calendar dates
	elapsed time	days
	Execution time	CPU clocks

GQM Approach /1

- Goal-Question-Metric (GQM) approach to process metrics provides a framework for deriving measures from organization or business goals.
- The "**Goal-Question-Metric**" (**GQM**) **approach** is a proven **method** for driving **goal**-oriented measures throughout a software organization. With **GQM**, we start by defining the **goals** we are trying to achieve, then clarifying the **questions** we are trying to answer with the data we collect.

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Basili's GQM

- Basili's GQM process model

Phase	Description
1	Developing a set of corporate, division and project goals
2	Generating questions that define those goals as completely as possible in a quantifiable way
3	Specifying measures (metrics) needed to be collected to answer those questions and to track process and product conformance to the goals
4	Developing mechanisms for data collection
5	Collecting, validating and analyzing the data in real time to provide feedback to projects for corrective action, to assess conformance to the goals and make recommendations for future improvements

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STTI-KL's GQM

- GQM process model defined by the Software Technology Transfer Initiative - Kaiserslautern (STTI-KL) in 1995 [Gresse 1995].

No.	Phase	Description
1	**Prestudy**	**The collection of information which is relevant to the introduction of a GQM-based measurement program**
2	**Identification of the GQM goals**	**Based on the prestudy, the GQM goals are derived from the informal organizational improvement goals and project goals**
3	**Production of GQM plan**	**Finding out the implicit knowledge of people with respect to the measurement goal by interviewing; Merge the results; Refine a GQM plan; Review the GQM plan**

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STTI-KL's GQM (cont'd)

No.	Phase	Description
4	Production of the measurement plan	Defining the data collection procedures and checking whether the prescribed data collection procedures are consistent with the project plan
5	Collection and validation of the data	Collecting measurement data according to the defined procedures; validated and stored data
6	Analysis of the data	Analyzing and interpreting of the collected data by feedback sessions
7	Packaging the experiences	Recording the results and experiences gained by the measurement program

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Solingen's GQM

- GQM process model defined by Solingen
  [Solingen et al. 1999]

No.	Phase	Procedures
1	Planning	Establish GQM team; Select improvement area; Select application project and establish a project team; Create project plan; Training and promotion
2	Definition	Define measurement goals; Review or produce software process models; Conduct GQM interviews; Define questions and hypotheses; Review questions and hypotheses; Define metrics; Check on metric consistency and completeness; Produce a GQM plan; Produce measurement plan; Produce analysis plan; Review plans

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Solingen's GQM (cont'd)

No.	Phase	Procedures
3	Data collection	Defining the data collection steps, forms, and using tools; Data collection start up and training; Building a measurement support system (MSS)
4	Interpretation	Preparation of a feedback session; Holding a feedback session; Reporting interpretations of measurement results; Cost and benefits analysis of a measurement program

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GQM Approach /2

- **Goal:** List major goals of development or maintenance project.
- **Question:** Derive from each goal the questions that must be answered to determine if the goals are being met.
- **Metrics:** Decide what must be measured in order to be able to answer the questions adequately.

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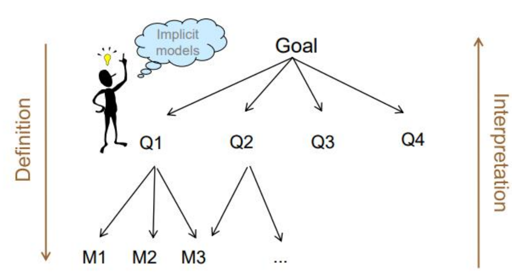
GQM Approach /3

GQM Core Elements

GQM has three elements:

- **Goals**
- **Questions (and associated Models)**
- **Measures**



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GQM Approach /3

- Goals
  - They define what the organization wants to improve
- Questions
  - They refine each goal to a more quantifiable way
- Metrics
  - They indicate the metrics required to answer each question

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GQM: Example /1

- Goal
  - Increase productivity
- Questions
- Metrics

Example from Fenton's Book

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GQM: Example /1

- Goal
  - Decrease development time
- Questions
- Metrics

Example from Fenton's Book

GQM: Example /1

GOAL: Evaluate effectiveness of coding standard

Example from Fenton's Book

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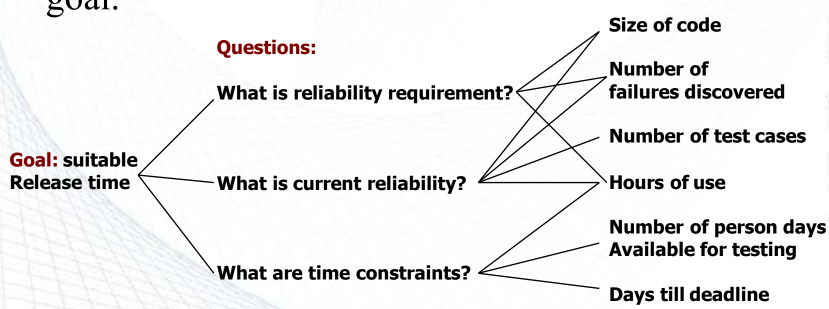
GQM: Example /2

- You are manager for a software development team and you have to decide upon the release time of your product. Construct a GQM tree related to this goal.



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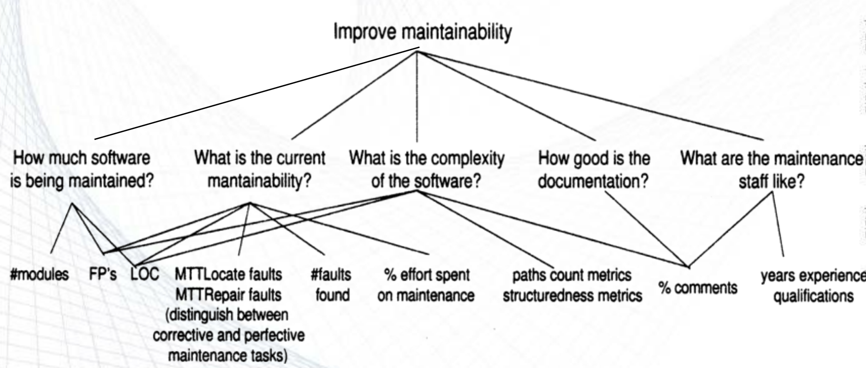
GQM: Example /3

- Construct a GQM graph for the goal of "improving maintainability of your developed software".



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Scenario

- Your organization has just completed a software process assessment.
- One of the findings was that your projects are delivering software functions and documentation on time, but customers are not as satisfied as you would like them to be.
- The related action item is: *Improve customer satisfaction*.
- Your process improvement team has identified this action item as one of its primary goals.
- You are a project manager, and you need a set of measures that will help your project make progress toward this goal.

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Entities managed by a project manager	Questions related to customer satisfaction
Inputs and resources	
people	Are our people qualified to produce the results the customer wants? Is personnel turnover hampering product quality?
subcontractors	Are the practices of our subcontractors consistent with those of the activities they support?
computers	Is the target system meeting its performance requirements? Is the target system reliable?
customer change requests	Do customer change requests contain the information that we must have to produce timely and effective changes?

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Entities managed by a project manager	Questions related to customer satisfaction
Products and by-products	
documents	Are the documents we produce readable?
	Is it possible to trace system features from one document to the next?
	Are documents concise and complete?
	Is the terminology correct?
source code & compiled products	Is the source code consistent with the documents?
	Is the source code error free?
	Does source code follow programming standards?
	Is the system response time adequate?
	Is the man-machine interface satisfactory?
plans	Are plans consistent with customer constraints?
	Are they kept up to date?
	Are plans and changes communicated to the customer?
budget	Are budgets consistent with plans?
	Are budgets consistent with customer constraints?

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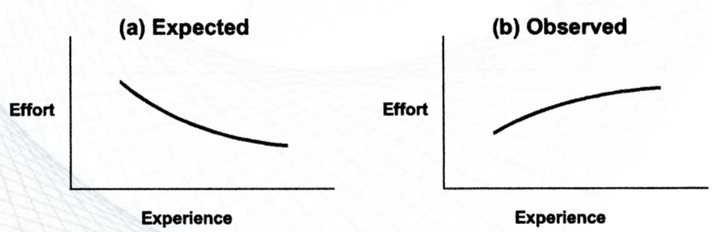
GQM: Data Analysis

- Suppose that you propose to collect data to examine the effects of programmer experience on development effort. You are expecting that effort (staff-hours) will decrease as the experience of the people assigned increases (Fig. a) but the collected data shows effort increasing with increasing experience (Fig. b). What may be wrong?



Technology factor? under-staffing? Cognitive overload? Rapidly changing requirements? staff morale? team-work problems?

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Case Study: AT&T's GQM

- Goal: Better code base inspection
- Subgoals:
  - Better inspection planning: [Plan]
  - Better monitor and control of the code: [Control]
  - Improving code inspection: [Improve]

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AT&T's GQM /1

GA: Code inspection: Plan

- **Q1:** How much does the inspection process cost?
- **M1-1:** Average effort per KLOC
- **M1-2:** Percentage of re-inspection
- **Q2:** How much calendar time does the inspection process take?
- **M2-1:** Average effort per KLOC
- **M2-2:** Total KLOC inspected

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AT&T's GQM /2

GB: Code inspection: Monitor & control

- **Q1:** What is the quality of the inspection software?
- **M1-1:** Average fault detected per KLOC
- **M1-2:** Average inspection rate
- **M1-3:** Average preparation rate
- **Q2:** To what degree did the staff conform to procedure?
- **M2-1:** Average inspection rate
- **M2-2:** Average preparation rate
- **M2-3:** Average lines of code inspected
- **M2-4:** Percentage of re-inspection
- **Q3:** What is the status of the inspection process?
- **M3:** Total KLOC inspected

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AT&T's GQM /3

GC: Code inspection: Improve

- **Q1:** How effective is the inspection process?
- **M1-1:** Defect removal efficiency
- **M1-2:** Average fault detected per KLOC
- **M1-3:** Average inspection rate
- **M1-4:** Average preparation rate
- **M1-5:** Average lines of code inspected
- **Q2:** What is the productivity of the inspection process?
- **M2-1:** Average effort per fault detected
- **M2-2:** Average inspection rate
- **M2-3:** Average preparation rate
- **M2-4:** Average lines of code inspected

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Case Study: HP's GQM

Three goals:

- **A:** Maximize customer satisfaction
- **B:** Minimize engineering effort and schedule
- **C:** Minimize defects

Example: HP's GQM-A /1

GA: Maximize customer satisfaction.

- **QA1:** What are the attributes of customer satisfaction?
  - **MA1:** Functionality, usability, reliability, performance, supportability.
- **QA2:** What are the key indicators of customer satisfaction?
  - **MA2:** Survey, quality function deployment (QFD).
- **QA3:** What aspects result in customer satisfaction?
  - **MA3:** Survey, QFD.
- **QA4:** How satisfied are customers?
  - **MA4:** Survey, interview record, number of customers severely affected by defects.
- **QA5:** How many customers are affected by a problem?
  - **MA5:** Number of duplicate defects by severity

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Example: HP's GQM-A /2

GA: Maximize customer satisfaction.

- **QA6:** How many problems are affecting the customer?
  - **MA6-1:** Incoming defect rate
  - **MA6-2:** Open critical and serious defects
  - **MA6-3:** Defect report/fix ratio
  - **MA6-4:** Post-release defect density
- **QA7:** How long does it take to fix a problem?
  - **MA7-1:** Mean time to acknowledge problem
  - **MA7-2:** Mean time to deliver solution
  - **MA7-3:** Scheduled vs. actual delivery
  - **MA7-4:** Customer expectation of time to fix

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Example: HP's GQM-A /2

GA: Maximize customer satisfaction.

- **QA8:** How does installing a fix affect the customer?
  - **MA8-1:** Time customers operation is down
  - **MA8-2:** Customers effort required during installation
- **QA9:** Where are the bottlenecks?
  - **MA9:** Backlog status, time spent doing different activities

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Example: HP's GQM-B /1

GB: Minimize engineering effort & schedule.

- **QB1:** Where are the worst rework loops in the process?
  - **MB1:** Person-months by product-component-activity.
- **QB2:** What are the total life-cycle maintenance and support costs for the product?
  - **MB2-1:** Person-months by product-component-activity.
  - **MB2-2:** Person-months by corrective, adaptive, perfective maintenance.
- **QB3:** What development methods affect maintenance costs?
  - **MB3:** Pre-release records of methods and post-release costs.

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Example: HP's GQM-B /2

GB: Minimize engineering effort & schedule.

- **QB4:** How maintainable is the product as changes occur?
  - **MB4-1:** Incoming problem rate
  - **MB4-2:** detect density
  - **MB4-3:** Code stability
  - **MB4-4:** Complexity
  - **MB4-5:** Number of modules changed to fix one detect
- **QB5:** What will process monitoring cost and where are the costs distributed?
  - **MB5:** Person-months and costs
- **QB6:** What will maintenance requirements be?
  - **MB6-1:** Code stability, complexity, size
  - **MB6-2:** Pre-release defect density

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Example: HP's GQM-B /3

GB: Minimize engineering effort & schedule.

- **QB7:** How can we predict cycle time, reliability, and effort?
  - **MB7-1:** Calendar time
  - **MB7-2:** Person-month
  - **MB7-3:** Defect density
  - **MB7-4:** Number of detects to fix
  - **MB7-5:** Defect report/fix ratio
  - **MB7-6:** Code stability
  - **MB7-7:** Complexity
  - **MB7-8:** Number of lines to change

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Example: HP's GQM-B /4

GB: Minimize engineering effort & schedule.

- **QB8:** What practices yield best results?
  - **MB8:** Correlations between pre-release practices and customer satisfaction data
- **QB9:** How much do the maintenance phase activities cost?
  - **MB9:** Personal-months and cost
- **QB10:** What are major cost components?
  - **MB10:** Person-months by product-component-activity
- **QB11:** How do costs change over time?
  - **MB11:** Track cost components over entity maintenance life- cycle.

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Example: HP's GQM-C /1

GC: Minimize defects

- **QC1:** What are key Indicators of process health and how are we doing?
  - **MC1:** Release schedule met, trends of defect density, serious and critical detects.
- **QC2:** What are high-leverage opportunities for preventive maintenance?
  - **MC2-1:** Defect categorization
  - **MC2-2:** Code stability
- **QC3:** Are fixes effective with less side effects?
  - **MC3:** Defect report/fix ratio

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Example: HP's GQM-C /2

GC: Minimize defects

- **QC4:** What is the post-release quality of each module?
  - **MC4:** Defect density, critical and serious detects.
- **QC5:** What are we doing right?
  - **MC5-1:** Defect removal efficiency
  - **MC5-2:** Defect report/fix ratio
- **QC6:** How do we know when to release?
  - **MC6-1:** Predicted defect detection and remaining defects.
  - **MC6-2:** Test coverage.

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Example: HP's GQM-C /3

GC: Minimize defects

- **QC7:** How effective is the development process in preventing defects?
  - **MC7:** Post-release detect density
- **QC8:** What can we predict will happen post-release based on pre-release data?
  - **MC8:** Corrections between pre-release complexity, defect density, stability, and customer survey data.
- **QC9:** What defects are getting through and there causes?
  - **MC9:** Detect categorization

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Exercise 4 :GQM/1

- Suppose that company X's main business goal is to "find a way to deal with fragile customer requirements" – that is, customer requirements that may be incomplete or that may change during the software development process. For this project, we want to create a measurement plan using GQM technique. During the step of "identify what we want to know or learn" we must specify the followings:
  - Persons affected
  - Products and by-products
  - Inputs and resources
  - Internal artifacts
  - Activities and flowpaths

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Exercise 4 :GQM/2

- Ask at least one question for each of the (a) - (e) that will contribute to the goal.

a) Persons affected: Who are the people that will be affected by this goal?
b) Products and by-products: What are the products covered by this goal?
c) Inputs and resources: What resources are potentially involved?
d) Internal artifacts: What are the artifacts affected?
e) Activities and flowpaths: What activities are involved?

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Exercise 4 :GQM/3

- Identify at least 3 relevant candidates for each of the (a) - (e) for this project.

a) Persons affected: Who are the people that will be affected by this goal?
   Customers; Project Managers; Programmers; Document personnel; Testers
b) Products and by-products: What are the products covered by this goal?
   Documents; Working system; Budget
c) Inputs and resources: What resources are potentially involved?
   People; Computers and development environment; Customers
d) Internal artifacts: What are the artifacts affected?
   Prototype applications; Change requests; Project history and previous metrics
e) Activities and flowpaths: What activities are involved?
   Developing; Requirements gathering; Acceptance testing

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Exercise /5

- Your organization has just completed a software process assessment. One of the findings was that your *projects are delivering software functions and documentation on time, but customers are not as satisfied as you would like them to be*.
- The process improvement team has identified ***"improving customer satisfaction"*** as one of its primary goals.
- You are a project manager, and you need a set of measures that will help your project make progress toward this goal.
- Devise subgoals to achieve this goal.

Start with asking questions about entities of interest (i.e., inputs and resources; internal artefacts; activities and flows; products and by-products)

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Entity-Question List (Part 1)

- **Inputs and Resources**

People	Are our people qualified to produce the results the customer wants? Is personnel turnover hampering product quality?
Subcontractors	Are the practices of our subcontractors consistent with those of the activities they support?
Computers	Is the target system meeting its performance requirements? Is the target system reliable?
Customer change requests	Do customer change requests contain the information that we must have to produce timely and effective changes?

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Entity-Question List (Part 2)

Internal Artifacts

Customer change requests (work in process)	How large is our backlog of customer change requests? Where are the backlogs occurring?

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Entity-Question List (Part 3)

Activities and Flowpaths

Development	Is development progress visible to the customer?
Testing	Are our testing procedures adequate for the operational use of the system? Does the customer accept the testing procedure and test results?
Fixing	Is the response time for fixing bugs compatible with customer constraints? Is change control adhered to? Are high-priority changes getting implemented in a timely fashion? Are status and progress visible to the customer?

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Entity-Question List (Part 4)

Products and By-products

Documents	Are the documents we produce readable? Is it possible to trace system features from one document to the next? Are documents concise and complete? Is the terminology correct?
Source code & compiled products	Is the source code consistent with the documents? Is the source code error free? Does source code follow programming standards? Is the system response time adequate? Is the man-machine interface satisfactory?
Plans	Are plans consistent with customer constraints? Are they kept up to date? Are plans and changes communicated to the customer?
Budget	Are budgets consistent with plans? Are budgets consistent with customer constraints?

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Grouping /1

Grouping #1 (documents)	Are the documents we produce readable? Is it possible to trace system features from one document to the next? Are documents concise and complete? Is the terminology correct?
Grouping #2 (software product)	Is the source code consistent with the documents? Is the source code error free? Does source code follow programming standards? Is the system response time adequate? Is the man-machine interface satisfactory? Is the target system meeting its performance requirements? Is the target system reliable? Is change control adhered to? Are testing procedures adequate for operational use of the system?

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Grouping /2

Grouping #3 (project management)	Are plans consistent with customer constraints? Are they kept up to date? Are budgets consistent with plans? Are budgets consistent with customer constraints?
Grouping #4 (change management)	Do customer change requests contain the information that we must have to produce timely and effective changes? How large is our backlog of customer change requests? Is the response time for fixing bugs compatible with customer constraints? Is change control adhered to? Are high-priority changes getting implemented in a timely fashion?

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Grouping /3

Grouping #5 (communications)	Are plans and changes communicated to the customer? Is development progress visible to the customer? Does the customer accept the testing procedure and test results? Are status and progress of change requests visible to the customer?
Other	Are our people qualified to produce the results the customer wants? Is personnel turnover hampering product quality? Are the practices of our subcontractors consistent with those of the activities they support?

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Derived Subgoals

Subgoal #1	Improve readability and traceability of documents.
Subgoal #2	Improve reliability and performance of released code.
Subgoal #3	Improve monitoring of plans and budgets.
Subgoal #4	Improve performance of the change management process.
Subgoal #5	Improve communications with the customer.
Subgoal #6	Assess qualification of the personnel.

#6 not directly related to the project manager

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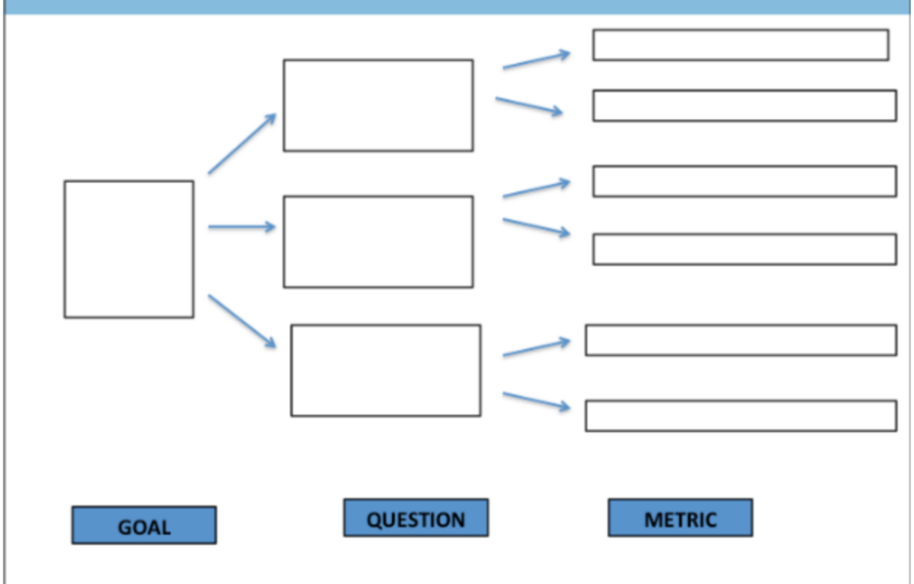
GQM Process



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